



Effects of surface characteristics of oxidized Zircaloy-4 on Pool boiling heat transfer

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ABSTRACT

Specimens of Zircaloy-4 were oxidized at 600 °C, 700 °C, and 800 °C for various time durations. Pool boiling experiments were conducted on these specimens to investigate the effects of oxidation on boiling heat transfer, which is essential for understanding cooling operations in nuclear reactors. Oxidation creates microstructural defects on the specimen surface, which enhances the surface wettability crucial for boiling heat transfer enhancement. However, the experimental results show that the critical heat flux (CHF) and heat transfer coefficient (HTC) values peaked at the specimens with oxide layer thickness between 20 and 30 μm, where the CHF increased by up to 147% for 600 °C, 168% for 700 °C, and 191% for 800 °C oxidized specimens compared to the bare specimen. This implies that surface wettability greatly affected heat transfer in specimens with oxide layer thicknesses below 30 μm, but not for higher thicknesses. The structural orientation of the cracks within the oxide layer have adverse effects on boiling heat transfer enhancement, where vertical cracks increase the surface wettability, but the horizontal cracks complicate bubble escape and obstruct the replenishment of the cooling fluid. The CHF ratio of specimens with oxide layer thicknesses exceeding 30 μm exhibited decreasing trend with decreasing thermal conductivity. As such, this study investigated the factors of the boiling heat transfer of oxidized Zircaloy-4, which are closely associated with thermal-hydraulic safety, therefore contributing to enhancement of the safety and efficiency of nuclear reactors.

1. Introduction

The fuel cladding in nuclear reactors is the first barrier against the release of radioactive fission products from the fuel rods into the reactor's coolant [1]. Therefore, cladding materials should possess high thermomechanical strength and excellent corrosion resistance to endure the operational environment inside a nuclear reactor core with consistent reliability [2]. In light-water reactors, Zircaloy-4 is currently widely used as the material for the fuel cladding due to its advantageous materialistic characteristics, some of which are minimal thermal neutron absorption, high ductility, and strong resistance to corrosion in high temperature water and steam conditions [3,4]. Other than the properties of the fuel cladding, efficient methods of cooling are also required for the structural integrity and safe operation of nuclear reactors [5].

Nucleate boiling is a highly effective heat transfer mechanism, utilizing the phase change from liquid to vapor [6,7]. It enables the transfer of large amounts of thermal energy with minimal temperature differences, due to the latent heat of vaporization, making it advantageous for high heat flux thermal management [8,9]. During the phase change process of nucleate boiling, vapor bubbles form, grow and depart from the heated surface [10]. The heat from the surface is absorbed by the vapor bubbles, and is carried away to the surrounding environment by the bubbles rising through buoyancy forces [11]. This mechanism enables the continuous replenishment of the cooling liquid, leading to efficient transfer of heat [12]. Pool boiling utilizes the characteristics of nucleate boiling heat transfer by submerging a heated component in a reservoir of cooling fluid [13]. This cooling method is widely used in nuclear reactors during severe accidents, where the reactor core is flooded and cooled by a pool of cooling fluid [14].

During the operation process of a nuclear reactor, the water coolant

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Nomenclature

CHF: critical heat flux.
 HTC: heat transfer coefficient.
 Sa: roughness, arithmetical mean height (μm).
 T: temperature measured at each point ($^{\circ}\text{C}$).
 x: distance (m).
 q'' : heat flux (W/m^2).
 k: thermal conductivity ($\text{W}/\text{m}\cdot\text{K}$).
 t: thickness of each layer in the specimen (m).
 U: uncertainty.
 h_{fg} : latent heat of evaporation (J/kg).
 g: gravitational acceleration (m/s^2).
 Δ : difference.
 ρ : density (kg/m^3).
 σ : surface tension (N/m).
 Subscripts.
 s: surface.
 super: superheat.
 sat: saturation.

absorbs heat from the fuel rods. A chemical reaction occurs between the fuel rod cladding and the water coolant, resulting in the development of an oxide layer [15]. This oxidation process induces the formation of microstructures on the surface [16]. In the previous studies concerning the surface roughness and surface wettability of oxidized Zircaloy-4, it was observed that increased oxidation time increased the oxide layer thickness, surface roughness and surface wettability [17,18]. This change of surface characteristic enhances boiling heat transfer by increasing the effective heat transfer area and promoting the capillary wicking effect of the cooling liquid towards the heated surface [19]. However, the oxide layer formed on the surface has reduced thermal conductivity compared to the base Zircaloy-4 material, and it may decrease the efficiency of heat transfer [20]. Therefore, the increase of the oxide layer thickness has both positive and negative effects on the boiling heat transfer of nuclear fuel claddings, and it is important to find the optimal oxide layer thickness and surface characteristics for the highest boiling heat transfer performance [21].

Investigating the effect of Zircaloy oxidation on pool boiling heat transfer is essential for understanding cooling operations in nuclear reactors. However, only a limited number of studies have addressed this issue. Kim et al. [19] observed that surface roughness of the oxidized Zircaloy affected the capillary wicking characteristics. The surface roughness and the water droplet spreading velocity were used to calculate the wicked liquid volume rate, which increased linearly with CHF enhancement. Lee et al. [22] observed that the CHF (critical heat flux) enhancement of the oxidized Zircaloy was 40% higher than that of the bare specimen, and the water contact angle had a close relationship with the CHF data. Lee et al. [23] compared the oxidation of silicide coated and uncoated Zircaloy and observed that the coating minimized surface oxidation. For boiling experiments, the surface roughness was considered as a key factor together with the contact angle, the CHF increasing steeply after a certain surface roughness. Yeom et al. [24] investigated the oxidation behavior of bare and coated ZIRLO and analyzed the surface morphology and chemistry, roughness, and static contact angle. Unlike the coated surfaces, the oxidation process had a significant effect on the surface wettability of bare surfaces. Although the degree of enhancement varied, all oxidized specimens exhibited CHF enhancement. He et al. [25] investigated the pool boiling heat transfer of oxidized Zircaloy variants, Zr-705, Zircaloy-4, and ZIRLO. It was observed that the HTC (heat transfer coefficient) of ZIRLO was greater than that of Zircaloy-4 and Zr-705, after nucleate boiling. Ahn et al. [26] achieved high surface wettability within a short period of time by

utilizing anodization, and observed that the multiscale structures on the surface mainly influenced the CHF enhancement at specimens with low contact angles.

An assessment of previous studies on the pool boiling of oxidized Zircaloy, arranged in Table 1, show that only a small variation of oxidized Zircaloy specimens with thin oxide layers below $10\text{ }\mu\text{m}$ were used, so there is a limit to understanding the boiling heat transfer performance of specimens with larger oxide layer thicknesses. The method of oxidation used on the Zircaloy specimen was mainly by ambient air or water, and at temperatures less than or equal to $700\text{ }^{\circ}\text{C}$. However, observations from the oxidation experiment show that cracks with distinct shapes (cracks with both horizontal and perpendicular directions) were mainly formed in the oxide layer at higher temperatures above $700\text{ }^{\circ}\text{C}$. Hence, further research is needed to study the oxidation effects of Zircaloy specimens with oxide layer thicknesses above $10\text{ }\mu\text{m}$, and effects of the cracks with distinct shapes formed at higher temperatures, may have on boiling heat transfer.

In this study, pool boiling experiments were performed using Zircaloy-4 specimens which were oxidized at temperatures of $600\text{ }^{\circ}\text{C}$, $700\text{ }^{\circ}\text{C}$, and $800\text{ }^{\circ}\text{C}$, in 80% humidified air. While recent studies on the boiling heat transfer of Zircaloy specimens have primarily focused on suppressing oxidation by applying oxidation-resistant coating materials [27–29], data from Table 1 indicate that the formation of an oxide layer with an appropriate thickness can enhance boiling heat transfer. In this context, this experiment aimed to observe the effect of surface characteristics of the oxidized Zircaloy-4 on pool boiling heat transfer performance, which is utilized in actual reactor emergency cooling operational conditions. It is also applicable for the operational cooling of nuclear reactors, as experimental studies on flow boiling have shown that surface modifications also contribute to CHF enhancement, in addition to the flow hydrodynamic parameters [30,31].

The CHF and HTC of the experimental results were used to evaluate the boiling heat transfer performance of each specimen. These factors of pool boiling measured and analyzed in this study heavily influence thermal-hydraulic safety and are the main criterion for the design of nuclear reactors. Therefore, investigating the effects of oxidized Zircaloy-4 surface characteristics on nucleate boiling heat transfer and identifying their optimal parameters, can contribute to enhancing the safety and efficiency of nuclear reactors under both emergency and operational cooling conditions.

2. Experiment

2.1. Oxidized Zircaloy-4 specimens

Flat Zircaloy-4 specimens were oxidized for various time periods in $600\text{ }^{\circ}\text{C}$, $700\text{ }^{\circ}\text{C}$, and $800\text{ }^{\circ}\text{C}$ temperature environments in a Lab House Tube Muffle Furnace (STF-80-3). Humidified air flow maintained at 80% humidity was introduced into the furnace chamber, to which the specimens were exposed. The oxidized Zircaloy-4 specimens possess dimensions of $25\text{ (length)} \times 25\text{ (width)} \times 0.5\text{ (height)}\text{ mm}$, and it was stored in a sealed plastic container to minimize additional oxidation. A specimen section with dimensions of $5\text{ (length)} \times 5\text{ (width)} \times 0.5\text{ (height)}\text{ mm}$, was cut and thermo-compressed in mounting media for Optical Microscopy (OM) measurement. To minimize errors associated with the cutting process, the oxide layer thickness was determined from a cross-section of the cut specimen section that had not been in contact with the cutting tool. The remaining specimen to be used for the boiling experiment possess dimensions of $25\text{ (length)} \times 20\text{ (width)} \times 0.5\text{ (height)}\text{ mm}$ after cutting, which thickness is identical to the fuel claddings actually used in nuclear reactors. The oxide layer is formed through metal consumption, not by material layer addition, and the total specimen thickness remains constant before and after oxidation.

The surface roughness of the specimen was measured using a Keyence VR-6000 non-contact 3D profiler, and the water droplet contact angle on its surface was measured using an optical tensiometer (Biolin

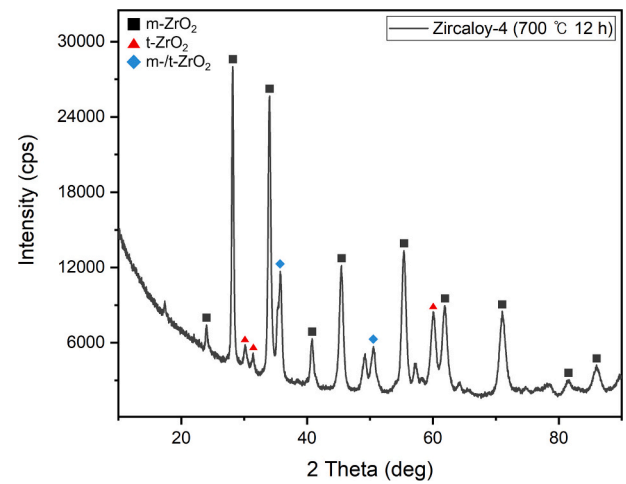
Table 1

Experimental data from previous studies.

Study (Pool boiling)	Oxidation type	Specimen type	Temperature [°C]	Maximum oxidation time [h]	Maximum oxide layer thickness [μm]	Maximum critical heat flux (CHF) [kW/m ²]
Kim et al. (2019) [19]	Air	Cr-coated Zircaloy-4	400	480	0.1	924.96
Lee et al. (2015) [22]	Air	Zircaloy-4	300, 450, 600	0.17	–	1800
Lee et al. (2019) [23]	Air	Zircaloy-4	700	1	4.43	1000
Yeom et al. (2020) [24]	Water	ZIRLO	360	360	1.3	879
He et al. (2022) [25]	–	Zr-705, Zircaloy-4, ZIRLO	–	–	9.745	584.55
Ahn et al. (2010) [26]	Anodization	Zircaloy-4	–	0.167	–	1924
Present study	Humidified air	Zircaloy-4	600, 700, 800	100	145.13	1777.49

Scientific Attenuation Theta Lite TL101). To minimize the effect of randomness, the oxide layer thickness, oxide surface roughness, and water droplet contact angle of the specimens were each measured on 5 separate points of the specimens, and the measured values were averaged to yield the final values presented in Table 2. One set of Zircaloy-4 specimens were oxidized under the same conditions for comparison and they exhibited near identical surface characteristic values, implying that the influence of their randomness is limited. Some oxidized specimens, which are not shown in Table 2, were excluded from the boiling experiment due to their deformations.

The air in which the Zircaloy-4 specimens were oxidized contains N₂ and O₂, so there is potential nitridation and formation of ZrN in the oxide layer. To address this issue, the possibility of nitridation was evaluated through X-ray diffraction (XRD) analysis on the oxide layer of the Zircaloy-4 specimen oxidized at 700 °C for 12 h (64 μm), and the results of the analysis are shown in Fig. 1. Only the characteristic diffraction peaks of monoclinic and tetragonal ZrO₂ were identified by the XRD analysis. Additionally, for the (111), (200), (220), (311), and (222) planes at $2\theta \approx 36.7^\circ$, 42.7° , 62.3° , 74.9° , and 89.9° , respectively, no signals attributable to ZrN were identified. Given the XRD phase identification limit of 3–5 wt%, the amount of ZrN in the oxide layer is estimated to be below this threshold, indicating no meaningful nitridation. In this context, it can be concluded that the oxide layers of the specimens used in the experiment consist entirely of oxides.

**Fig. 1.** X-ray diffraction (XRD) pattern of the Zircaloy-4 oxide layer.

2.2. Experimental facility setup

A schematic diagram of the experimental facility is shown in Fig. 2.

Table 2

Oxide layer thickness, surface roughness, and water droplet contact angle of the oxidized Zircaloy-4.

600 °C			700 °C			800 °C		
Oxide layer thickness	Oxide surface roughness (Sa)	Water droplet contact angle	Oxide layer thickness	Oxide surface roughness (Sa)	Water droplet contact angle	Oxide layer thickness	Oxide surface roughness (Sa)	Water droplet contact angle
11.56 ± 3.81 μm	1.114 ± 0.16 μm	23.8 ± 1.4°	14.22 ± 4.49 μm	1.251 ± 0.158 μm	15.8 ± 2.4°	15.65 ± 2.86 μm	1.607 ± 0.131 μm	36.6 ± 3.5°
23.21 ± 4.19 μm	1.097 ± 0.195 μm	20.9 ± 4.5°	27.14 ± 3.12 μm	1.891 ± 0.189 μm	12.4 ± 2.1°	25.61 ± 5.14 μm	1.546 ± 0.165 μm	26.2 ± 4.3°
30.68 ± 4.23 μm	1.179 ± 0.101 μm	19.6 ± 5.4°	41.82 ± 5.26 μm	2.322 ± 0.215 μm	8.6 ± 1.1°	30.58 ± 7.86 μm	1.88 ± 0.058 μm	24.1 ± 2.3°
39.57 ± 2.95 μm	1.502 ± 0.301 μm	17.6 ± 4.4°	51.27 ± 4.87 μm	3.204 ± 0.299 μm	4.3 ± 1.7°	33.63 ± 6.92 μm	2.126 ± 0.189 μm	22.1 ± 3.4°
51.55 ± 3.65 μm	1.327 ± 0.321 μm	16.7 ± 4.3°	66.42 ± 4.36 μm	4.909 ± 0.351 μm	0°	47.24 ± 7.31 μm	2.281 ± 0.314 μm	18.2 ± 1.8°
60.22 ± 2.84 μm	1.457 ± 0.386 μm	5.1 ± 1.8°	78.09 ± 3.89 μm	5.261 ± 0.271 μm	0°	51.54 ± 6.32 μm	3.188 ± 0.102 μm	0°
70.39 ± 2.56 μm	2.674 ± 0.292 μm	0°	92.69 ± 5.72 μm	5.476 ± 0.347 μm	0°	65.75 ± 5.98 μm	2.983 ± 0.354 μm	0°
77.27 ± 3.49 μm	2.866 ± 0.543 μm	0°	145.13 ± 6.45 μm	15.109 ± 0.862 μm	0°	73.77 ± 6.74 μm	5.211 ± 0.234 μm	0°
90.11 ± 4.17 μm	5.779 ± 0.413 μm	0°						
101.71 ± 5.75 μm	10.232 ± 0.873 μm	0°						

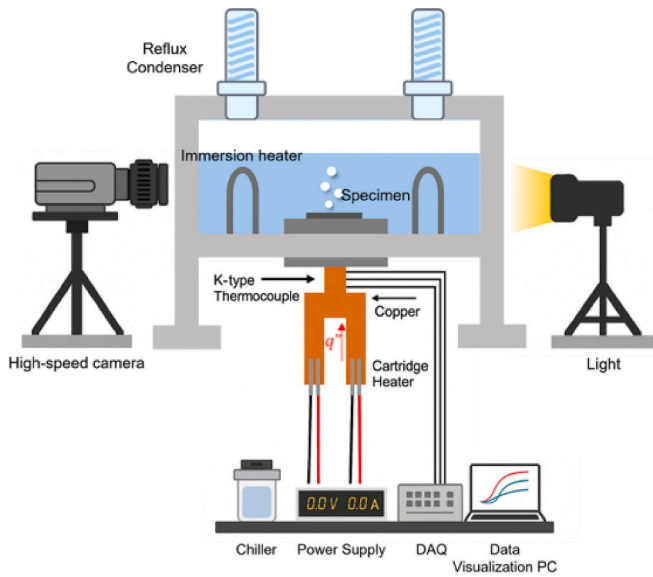


Fig. 2. Schematic of the experimental facility.

The experimental facility largely consists of the pool boiling tank, immersion heaters, reflux condensers, and the test section. The pool boiling tank, with dimensions of 350 (length) $\times$ 150 (width) $\times$ 200 (height) mm and made of aluminum, is the section where the pool boiling experiment is conducted. Both the front and rear walls fitted onto the tank are made of transparent polycarbonate panels, allowing visual observation and recording of the experimental process. To heat up and sustain the temperature of the cooling fluid at its saturation point, two immersion heaters were installed at the bottom of the tank. A constant level of the cooling fluid within the tank is achieved by installing a reflux condenser at the top, which condenses rising vapor and returns the liquid back to the tank. A water chiller connected to the reflux condenser regulates a bath of cooling water at a constant temperature, and cycles the cooling water within the condensing system, ensuring its performance. During the experiment, turbulent flows of the cooling fluid in the tank, induced by immersion heaters, can interfere with the boiling behavior at the test section, potentially compromising experimental results. To reduce such turbulent flows, polycarbonate barriers are installed between the heaters and the test section to ensure undisturbed experimental measurement. A high-speed camera (Photron FASTCAM) is used to capture images from the experiment at 13,000 fps with a resolution of 512×512 pixels.

A detailed schematic of the test section is shown in Fig. 3. The oxidized Zircaloy-4 specimen is secured on a PEEK jig by using a thermally conductive binary epoxy adhesive (Duralco 4460), preventing leakage of the cooling fluid into the test section. The dimensions of the specimen, which is 25 (length) $\times$ 20 (width) mm, were set larger than that of the copper block, which is 15 (length) $\times$ 15 (width) mm, to ensure complete coverage of the heated area. The heat flux was calculated using the area of the specimen in contact with the copper block heater, and the overhanging portion is thermally inactive and does not contribute to the calculation. Four DC cartridge heaters, each with a maximum heating power of 240 W, are embedded into the bottom part of the copper block, with two cartridge heaters positioned on each side. These cartridge heaters are powered by two DC power supplies (Ametek XG 60–25), with each power supply connected to a pair of cartridge heaters. Heat supplied from the cartridge heaters is transferred to the specimen through the copper block by conduction.

The temperature of the copper block throughout the experiment is measured at three specific locations using three K-type thermocouples (Watlow AFEC0HQ060G4030) inserted inside the copper block. These temperatures are used to calculate the surface temperature and heat flux

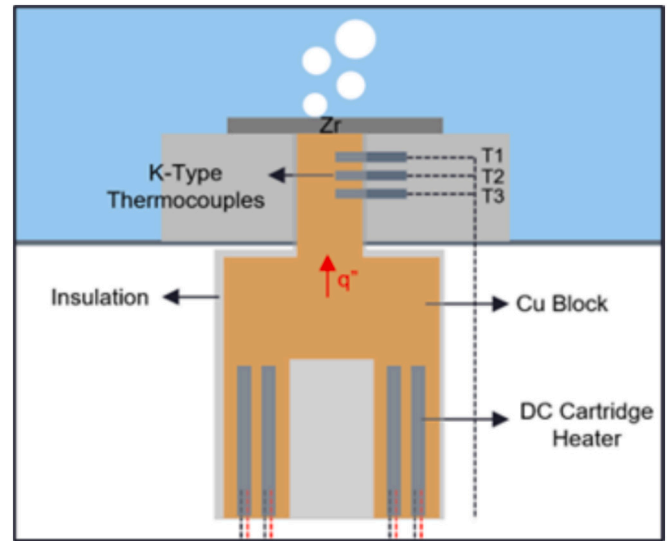


Fig. 3. Detailed schematic of the test section.

of the specimen. The copper block heater was positioned beneath the Zircaloy-4 specimen, and mechanically secured using support jacks to ensure stable surface contact. Thermal grease (Thermal Grizzly Kryonaut Extreme) was applied between the copper block and the specimen to minimize interfacial thermal resistance and errors. The bottom part of the copper block is thermally insulated from the surrounding environment with fiberglass to ensure experimental accuracy. Data from the experiment, such as applied voltage, current, and measured temperature, are obtained through a data acquisition system (Keysight DAQ970A) at 1 Hz frequency, enabling comprehensive data analysis.

2.3. Experimental procedures

For the pool boiling experiments, deionized water was used as the cooling fluid. The cooling fluid in the pool boiling tank was preheated by the two immersion heaters for an hour to remove noncondensable gases, ensuring optimal experimental conditions. After preheating the cooling fluid, 5 V voltage was applied from each power supply to the cartridge heaters, with the purpose of setting the heat flux at the specimen surface to near zero. When the surface temperature and heat flux of the specimen were stabilized, the applied voltage from each power supply was increased in increments of 1 V every 15 min for the purpose of observing the onset of nucleate boiling (ONB), after which the applied voltage was increased in increments of 2 V every 15 min until the critical heat flux (CHF) point. In an incremental step of 15 min, the surface temperature and heat flux increased for three minutes, and stabilized for 12 min. The CHF value, which indicates the maximum heat flux boundary for safe operational conditions, was identified at the point just before the surface temperature exhibited a rapid increase due to the formation of vapor films on the surface.

2.4. Data analysis and experimental uncertainties

In this study, the surface temperature of the specimen was estimated based on temperature measurements from three K-type thermocouples (1 mm in diameter) embedded within the copper block. The thermocouples were inserted at three collinear locations, each spaced 3 mm apart along a straight line perpendicular to the surface of the copper block, where the first thermocouple was positioned 3 mm below the surface.

To estimate the temperature gradient near the surface, the second order backward finite difference method was applied. The temperature gradient at the copper block was calculated using the following equa-

tion:

$$\frac{dT}{dx} = \frac{3T_1 - 4T_2 + T_3}{2\Delta x} \quad (1)$$

where x is the distance between two consecutive locations of the thermocouples, which is 3 mm. T_1 , T_2 , and T_3 are the temperatures measured at depths of 3 mm, 6 mm, and 9 mm from the surface of the copper block, respectively.

The heat flux applied to the specimen was calculated by applying the temperature gradient at the copper block to the Fourier's law of thermal conduction in one dimension:

$$q'' = -k \frac{dT}{dx} \quad (2)$$

where k is the thermal conductivity of the copper block (385 W/m·K).

The temperature at the surface of the oxidized Zircaloy-4 specimen in contact with the cooling fluid was also calculated using the Fourier's law of thermal conduction between the specimen surface and the location of the first thermocouple:

$$T_s = T_1 - q'' \left(\frac{t_{Cu}}{k_{Cu}} + \frac{t_{tg}}{k_{tg}} + \frac{t_{Zr}}{k_{Zr}} + 2 \frac{t_{ZrO2}}{k_{ZrO2}} \right) \quad (3)$$

where T_s is the temperature at the surface of the specimen, T_1 is the temperature measured by the first thermocouple, q'' is the heat flux of the copper block. t_{Cu} , t_{tg} , t_{Zr} , t_{ZrO2} are the thicknesses of the copper block (3 mm), thermal grease (0.1 mm), Zircaloy-4, and the oxide layer of Zircaloy-4, respectively. k_{Cu} , k_{tg} , k_{Zr} , k_{ZrO2} are the thermal conductivities of the copper block (385 W/m·K), thermal grease (14.2 W/m·K), Zircaloy-4 (16.7 W/m·K), and the oxide layer of Zircaloy-4 (1.5 W/m·K), respectively. The Zircaloy-4 specimen was oxidized on both front and back surfaces, hence twice of the oxide layer thickness variable is used in the equation.

The wall superheat was derived from the difference between the temperature at the surface of the specimen (T_s) and the saturation temperature (T_{sat}) of deionized water (100 °C):

$$\Delta T_{super} = T_s - T_{sat} \quad (4)$$

The heat transfer coefficient (HTC) of the specimen was calculated using Newton's law of cooling:

$$HTC = \frac{q''}{\Delta T_{super}} \quad (5)$$

where q'' is the heat flux and ΔT_{super} is the wall superheat.

The uncertainty of the thermocouples was ± 0.2 K, and the uncertainty of the distance between neighboring thermocouples is 0.01 mm. The uncertainties of the heat flux and wall superheat were estimated from the following equations, which were used in the study by Ahn et al. [26] and were based on the research by Kline and McClintock [32]:

$$\frac{U_{q''}}{q''} = \sqrt{\left(\frac{U_{T_2-T_1}}{T_2 - T_1} \right)^2 + \left(\frac{U_{\Delta x}}{\Delta x} \right)^2} \quad (6)$$

$$\frac{U_{\Delta T_{super}}}{\Delta T_{super}} = \sqrt{\left(\frac{U_{T_1}}{T_1} \right)^2 + \left(\frac{U_{q''}}{q''} \right)^2 + \left(\frac{U_{\Delta x}}{\Delta x} \right)^2} \quad (7)$$

where $U_{q''}$ is the uncertainty of the heat flux, $U_{T_2-T_1}$ is the uncertainty of the difference between temperatures measured at neighboring thermocouples, $U_{\Delta x}$ is the uncertainty of the distance between these thermocouples, $U_{\Delta T_{super}}$ is the uncertainty of the wall superheat. The uncertainty of the heat flux was estimated to be 10.1% at the heat flux of 100 kW/m², and 1.6% at 900 kW/m². The uncertainty of the wall superheat was estimated to be 11.0% at the heat flux of 100 kW/m², and 2.2% at 900

kW/m². The uncertainty of the parameters were presented in Table 3.

3. Experimental results

3.1. Boiling curves

Fig. 4 shows the boiling curves of the oxidized Zircaloy-4 specimens, where the x-axis of the graphs represents the wall superheat, while the y-axis represents the heat flux. The CHF of the bare Zircaloy-4 was 932.16 kW/m², which was denoted in all three graphs for relative comparison. This CHF value of the bare specimen was similar with the reference value of 963.85 kW/m², which was derived from Zuber's CHF correlation [33], indicating that the experimental results have good agreement with theoretical predictions.

Zuber's CHF correlation is based on hydrodynamic instability of vapor columns on smooth and uniformly wetted surfaces. In this context, the parameters in the correlation are limited to fluid properties and gravity effects [33]. On the other hand, oxidized surfaces deviate from hydrodynamic instabilities on smooth surfaces, because surface characteristics such as cracks and protrusions alter the vapor bubble escape and liquid replenishment dynamics near the surface [34]. Therefore, the CHF is no longer governed by hydrodynamic instability of vapor columns, so Zuber's correlation is not applicable for oxidized surfaces, highlighting the necessity of experimental investigations.

$$q''_{Zuber} = 0.131 h_{lg} \rho_{lg}^{0.5} [\sigma g (\rho_l - \rho_g)]^{0.25} \quad (8)$$

The repetitive experimental results of the bare specimens also show good agreement, with the error rate of the CHF between experiments within 10%. The reliability of the experimental results was further supported by the dense distribution of oxide layer thicknesses, where the thickness interval between specimens was approximately 10 μm, enabling systematic evaluation over a broad range of oxide layer thicknesses. This dense dataset reduced the possibility that the conclusions were affected by a single anomalous result.

Fig. 4 (a) shows the boiling curves of Zircaloy-4 specimens oxidized at 600 °C, with oxide layer thicknesses ranging from 11 μm to 101 μm. The graph shows that the CHF peaked at the value of 1373.57 kW/m² when the oxide layer thickness is 23 μm, which is an increase of 47.35% compared to the bare specimen. The lowest CHF value was 571.16 kW/m² when the oxide layer thickness is 61 μm, which is a 38.73% decrease. Fig. 4 (b) shows the boiling curves of Zircaloy-4 specimens oxidized at 700 °C, with oxide layer thicknesses ranging from 14 μm to 145 μm. The graph shows that the CHF peaked at the value of 1572.10 kW/m² when the oxide layer thickness is 27 μm, which is an increase of 68.65% compared to the bare specimen. The lowest CHF value of 541.11 kW/m² when the oxide layer thickness is 145 μm, which is a 41.95% decrease. Fig. 4 (c) shows the boiling curves of Zircaloy-4 specimens oxidized at 800 °C, with oxide layer thicknesses ranging from 15 μm to 74 μm. The graph shows that the CHF peaked at the value of 1777.49 kW/m² when the oxide layer thickness is 25 μm, which is an increase of 90.68% compared to the bare specimen. The lowest CHF value was 871.14 kW/m²

Table 3
Uncertainty of the parameters.

Quantity	Symbol	Value
Uncertainty of the temperature measured at the thermocouple	U_T	± 0.2 K
Uncertainty of the distance between neighboring thermocouples	$U_{\Delta x}$	0.01 mm
Uncertainty of the difference between temperatures measured at neighboring thermocouples	$U_{T_2-T_1}$	± 0.283 K
Uncertainty of the heat flux at the heat flux of 100 kW/m ² and 900 kW/m ²	$\frac{U_{q''}}{q''}$	10.1%, 11.0%
Uncertainty of the wall superheat at the heat flux of 100 kW/m ² and 900 kW/m ²	$\frac{U_{\Delta T_{super}}}{\Delta T_{super}}$	1.6%, 2.2%

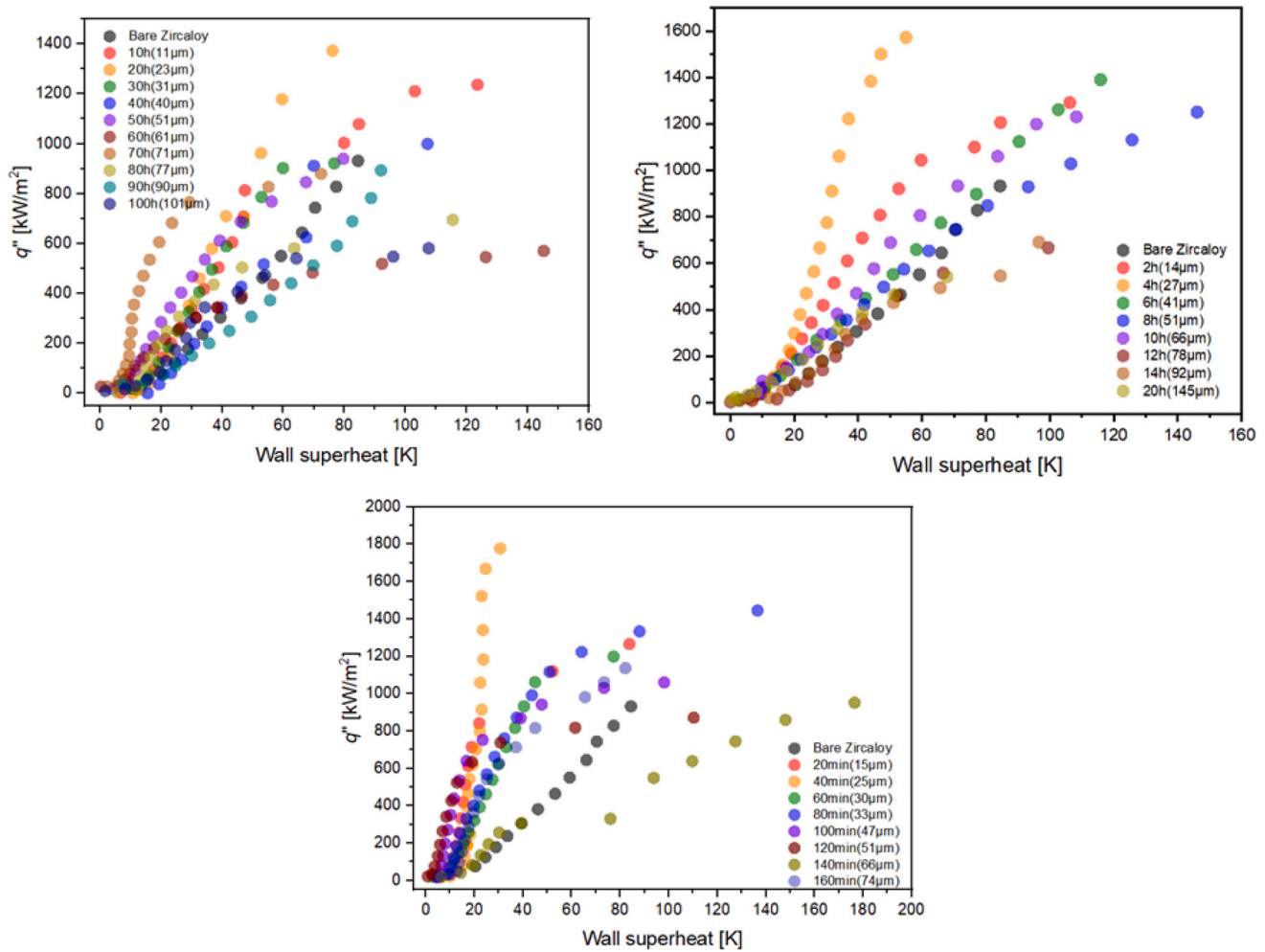


Fig. 4. (a) Boiling curves of Zircaloy-4 specimens oxidized at 600 °C. (b) Boiling curves of Zircaloy-4 specimens oxidized at 700 °C. (c) Boiling curves of Zircaloy-4 specimens oxidized at 800 °C.

m^2 when the oxide layer thickness is 51 μm , which is a 6.55% decrease.

Fig. 5 shows the CHF enhancement of the oxidized Zircaloy-4 specimens compared to the bare specimen. The graph shows that for all temperatures, the maximum CHF enhancement was at the oxide layer

thicknesses between 20 μm and 30 μm . After the maximum point, the CHF enhancement shows a generally decreasing trend for all temperatures. The CHF value becomes similar with the bare specimen at the oxide layer thickness of 31 μm for 600 °C, between 66 μm and 78 μm for 700 °C, and between 47 μm and 51 μm for 800 °C. The CHF enhancement was generally larger for 700 °C and 800 °C, than for 600 °C.

3.2. Heat transfer coefficient graphs

Fig. 6 shows the heat transfer coefficient (HTC) graphs of the oxidized Zircaloy-4 specimens, where the x-axis of the graphs represents the heat flux, while the y-axis represents the HTC. The maximum HTC of the bare Zircaloy-4 was 11.04 $\text{kW}/\text{m}^2\text{K}$, which is similar to the results from the previous studies, and it was denoted in all three graphs for relative comparison. The graphs show that the maximum HTC was 33.21 $\text{kW}/\text{m}^2\text{K}$ for 600 °C at 71 μm oxide layer thickness, 32.98 $\text{kW}/\text{m}^2\text{K}$ for 700 °C at 27 μm , and 67.89 $\text{kW}/\text{m}^2\text{K}$ for 800 °C at 25 μm . For 700 °C and 800 °C, the maximum HTC was at the oxide layer thicknesses between 20 μm and 30 μm , where the maximum CHF for both temperatures occurred. However, for 600 °C the maximum HTC was at the higher oxide layer thickness of 71 μm , rather than the thickness where the maximum CHF occurred. This phenomenon where higher wettability does not automatically relate to higher HTC, might be attributed to the distinct surface characteristics of the specimens, which affects the bubble dynamics. The HTC of the oxidized Zircaloy-4 specimens was generally higher for 800 °C, compared to 600 °C and 700 °C.

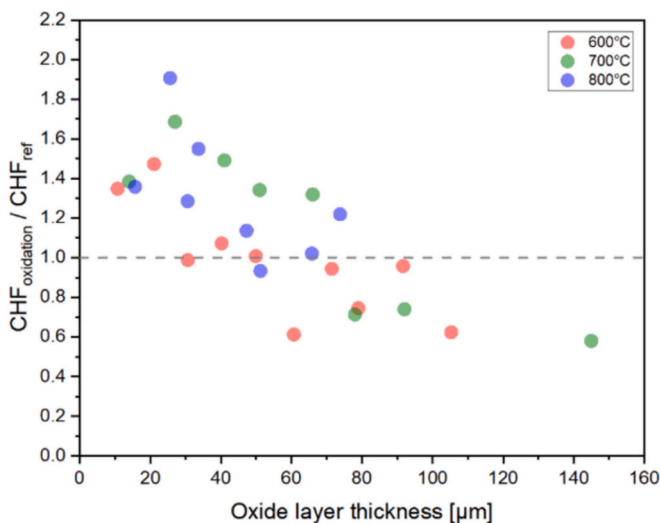


Fig. 5. CHF enhancement of Zircaloy-4 specimens oxidized at various temperatures.

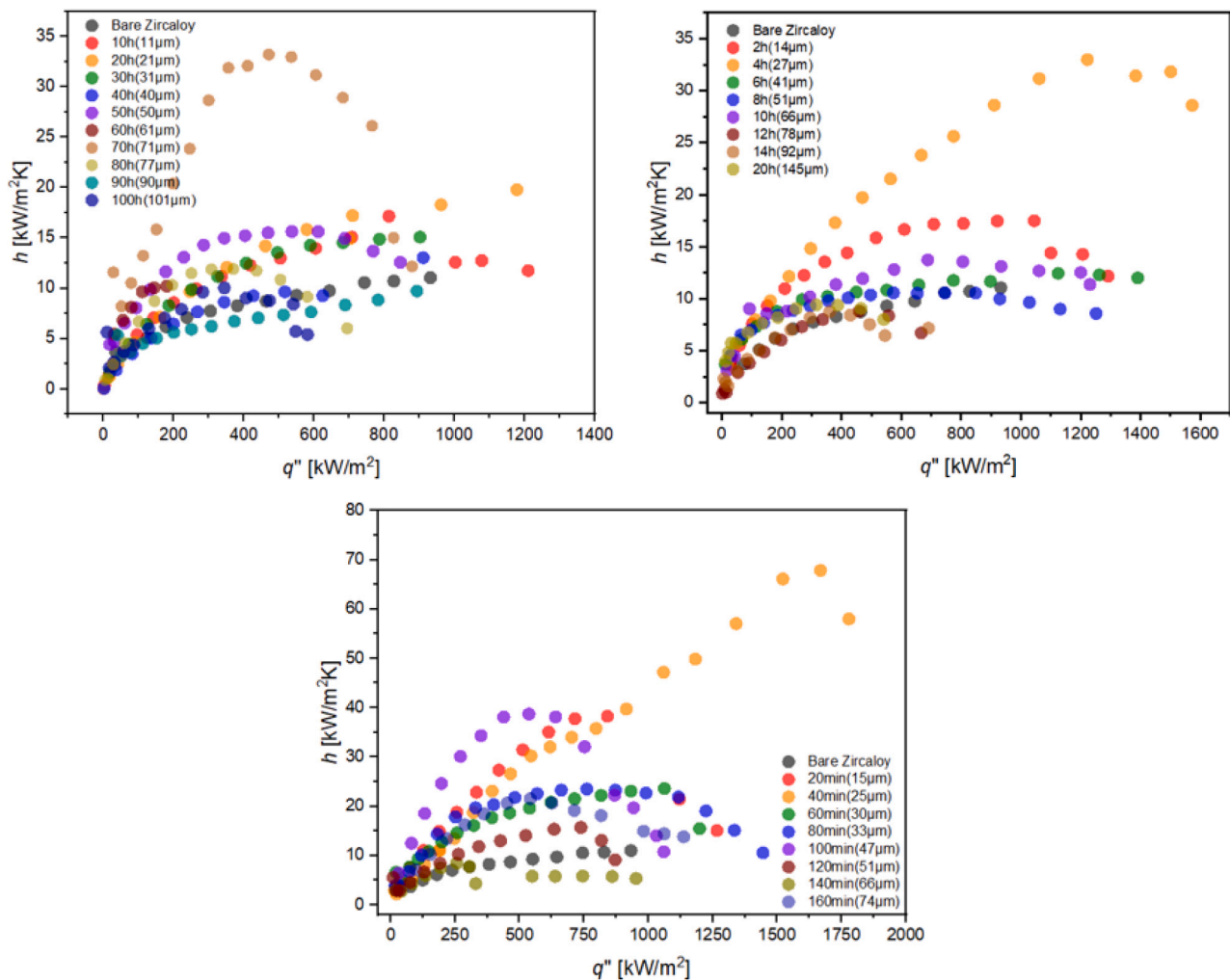


Fig. 6. (a) HTC of Zircaloy-4 specimens oxidized at 600 °C. (b) HTC of Zircaloy-4 specimens oxidized at 700 °C. (c) HTC of Zircaloy-4 specimens oxidized at 800 °C.

4. Discussion

4.1. Effect of surface characteristics

As shown in Table 2, the surface roughness increased with the oxide layer thickness. During oxidation, compressive stresses are generated at the oxide–metal interface, as the molar volume of Zircaloy-4 oxide (21.69 cm^3/mol) is larger than that of the original Zircaloy-4 (13.97 cm^3/mol) [35]. As the oxide layer grows with the oxidation time, the original structure can no longer sustain these stresses, leading to the formation of cracks [36]. This stress-relief mechanism results in visible microstructural defects at the surface. When in contact with water, these microstructures induce the capillary wicking effect, which facilitates the movement of water towards the surface [37]. This makes the oxidized Zircaloy-4 surface hydrophilic. During pool boiling, this phenomenon prevents the formation of hot spots and vapor films on the surface, delaying the onset of CHF, thus enhancing heat transfer performance [38].

It was also observed that the surface wettability increased with the surface roughness, aligning with the Wenzel model. However, contrary to the assumption that higher surface wettability will lead to higher heat transfer performance, the results from the pool boiling experiments showed that the highest CHF and HTC values were focused at oxide layer thicknesses from 20 to 30 μm . This shows that while surface wettability greatly affected heat transfer performance in specimens with oxide layer thicknesses below 30 μm , it was less the case for specimens with higher

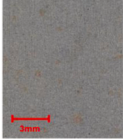
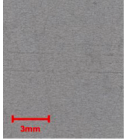
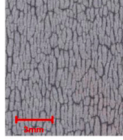
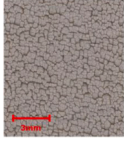
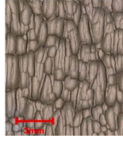
oxide layer thicknesses.

Specimens oxidized at 800 °C exhibit lower crack density, surface roughness and wettability, yet develop noticeable surface protrusions and wider, more prominent cracks, as shown in Table 4. Despite their lower surface roughness and wettability, these specimens demonstrate relatively higher CHF and HTC values compared to those oxidized at other temperatures. This suggests that, across different oxidation temperatures, surface characteristics other than roughness and wettability, such as crack morphology and surface topology, have a more significant influence on boiling heat transfer performance. From the high-speed images of the bubble dynamics taken from the pool boiling experiments in Table 5, it can be observed that the generation of bubbles on the surface was weaker for the 600 °C oxidized specimen than for the 800 °C oxidized specimen, hence the lower heat fluxes measured.

4.2. Effect of crack structure

During boiling heat transfer, vapor bubbles absorb heat from the surface through phase change in the form of latent heat, so rapid bubble generation and removal is crucial for efficient boiling heat transfer performance. As mentioned in the previous section, forming defects on the specimen surface such as oxidation is a method to increase bubble generation. However, the structural orientation of these cracks have two-sided effects on boiling heat transfer enhancement, as depicted in Fig. 7. Vertical cracks act like tubes which induce the capillary wicking effect and draw the cooling fluid towards the heated surface. On the

Table 4
Surface images of specific specimens.

Oxidation temperature	600 °C	700 °C	800 °C
Surface images			
Oxide layer thickness	$30.68 \pm 4.23 \mu\text{m}$	$27.14 \pm 3.12 \mu\text{m}$	$30.58 \pm 7.86 \mu\text{m}$
Oxide surface roughness (Sa)	$1.179 \pm 0.101 \mu\text{m}$	$1.891 \pm 0.189 \mu\text{m}$	$1.88 \pm 0.058 \mu\text{m}$
Water droplet contact angle	$19.6 \pm 5.4^\circ$	$12.4 \pm 2.1^\circ$	$24.1 \pm 2.3^\circ$
Surface images			
Oxide layer thickness	$77.27 \pm 3.49 \mu\text{m}$	$78.09 \pm 3.89 \mu\text{m}$	$73.77 \pm 6.74 \mu\text{m}$
Oxide surface roughness (Sa)	$2.866 \pm 0.543 \mu\text{m}$	$5.261 \pm 0.271 \mu\text{m}$	$5.211 \pm 0.234 \mu\text{m}$
Water droplet contact angle	0°	0°	0°

other hand, horizontal cracks complicate the escape route of the bubbles, making the bubble escape route longer than in vertical cracks.

In the previous study by Akbari et al. [39], porous foam covers with layered cells were used in pool boiling experiments. It was observed that the larger bubbles generated inside the porous structure failed to move up through the narrow openings to the upper cells, and travelled horizontally for a longer distance until they reached the larger openings located at the sides. This phenomenon obstructed replenishment of the cooling fluid inside the porous structure, which in turn hindered boiling heat transfer performance. Therefore, ensuring that the bubble escape route is as short as possible during boiling heat transfer, facilitates surface heat removal and cooling fluid replenishment.

Optical Microscope (OM) cross-section images in Table 6 show that while only vertical cracks were observed for 600 °C, both vertical and horizontal cracks were observed at 700 °C and 800 °C, being the most prominent at 700 °C. The cracks in the horizontal direction, which diverged from the vertical cracks exposed to the surface, were more developed for specimens with higher oxide layer thicknesses. This characteristic is one of the driving forces behind the deterioration of heat transfer performance at high oxide layer thicknesses for specimens oxidized at the same temperature. On the other hand, it was not the case when comparing between specimens oxidized at different temperatures,

indicating that other factors were more decisive between such conditions.

Snapshots of the experimental high-speed images of various specimens, in time intervals of 8.8 ms were presented in Table 7. The following specimens were analyzed: 600 °C 80 h specimen (oxide layer thickness 77.27 μm) at the heat flux of 27.75 kW/m^2 , 800 °C 120 min specimen (oxide layer thickness 51.54 μm) at the heat flux of 34.09 kW/m^2 and the 800 °C 160 min specimen (oxide layer thickness 73.77 μm) at the heat flux of 31.78 kW/m^2 . Table 7 shows that the approximate bubble departure time of the vapor bubbles was the largest in the 600 °C 80 h specimen (around 44.00 ms), followed by the 800 °C 160 min specimen (around 35.2 ms), and lastly the 800 °C 120 min specimen (around 26.4 ms).

When comparing specimens with similar oxide thicknesses, the bubble departure time of the 600 °C specimen was larger than that of the 800 °C specimen, despite the 800 °C specimen having horizontal cracks that complicate the bubble escape route. This may be attributed to the fact that the cracks on the 600 °C specimen were significantly narrower (below 10 μm) than that of the 800 °C specimen (maximum width around 42.5 μm), as shown in Table 6. Narrower cracks generate higher capillary forces in theory, but they also have a higher tendency to merge and hinder bubble escape, so their effect on boiling heat transfer depends on whether they provide effective liquid replenishment without blocking vapor removal. When comparing specimens oxidized at 800 °C, the bubble departure time of the 800 °C 160 min specimen was larger than that of the 800 °C 120 min specimen. Specimens with higher thicknesses have more developed horizontal cracks, reinforcing the analysis of the manuscript that horizontal cracks complicate bubble escape.

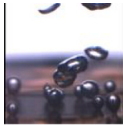
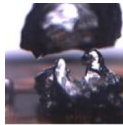
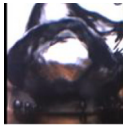
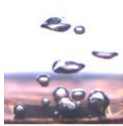
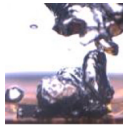
4.3. Effect of thermal conductivity

During the pool boiling experiment, the heat flux was applied through a heated copper block placed beneath the specimen. Therefore, the specimen's thermal conductivity and conduction resistance mainly affects boiling heat transfer performance by defining how much of the heat applied by the copper block actually reaches the boiling surface. The thermal conductivity of oxidized Zircaloy-4 (1.5 $\text{W/m}\cdot\text{K}$), is significantly smaller than that of the original Zircaloy-4 (16.7 $\text{W/m}\cdot\text{K}$), drastically decreasing the overall thermal conductivity of oxidized Zircaloy-4 specimens with thicker oxide layers. The total thermal conductivity of the specimen was derived by using the following equation for thermal resistances in series:

$$k_{\text{eff}} = \frac{t_{\text{total}}}{\frac{t_{\text{Zr}}}{k_{\text{Zr}}} + \frac{2t_{\text{ZrO}_2}}{k_{\text{ZrO}_2}}} \quad (9)$$

where k_{eff} , k_{Zr} , k_{ZrO_2} are the thermal conductivities of the specimen, Zircaloy-4 (16.7 $\text{W/m}\cdot\text{K}$), and the oxide layer of Zircaloy-4 (1.5 $\text{W/m}\cdot\text{K}$), respectively. t_{total} , t_{Zr} , t_{ZrO_2} are the total thickness of the specimen (0.5

Table 5
High-speed images of the bubble dynamics of different specimens.

		Heat flux (kW/m^2)			
Oxidized at 600 °C for 80 h	High-speed images	12.67	145.25	370.24	695.00
					
Oxidized at 800 °C for 120 min	High-speed images	34.09	262.44	634.01	871.14
					

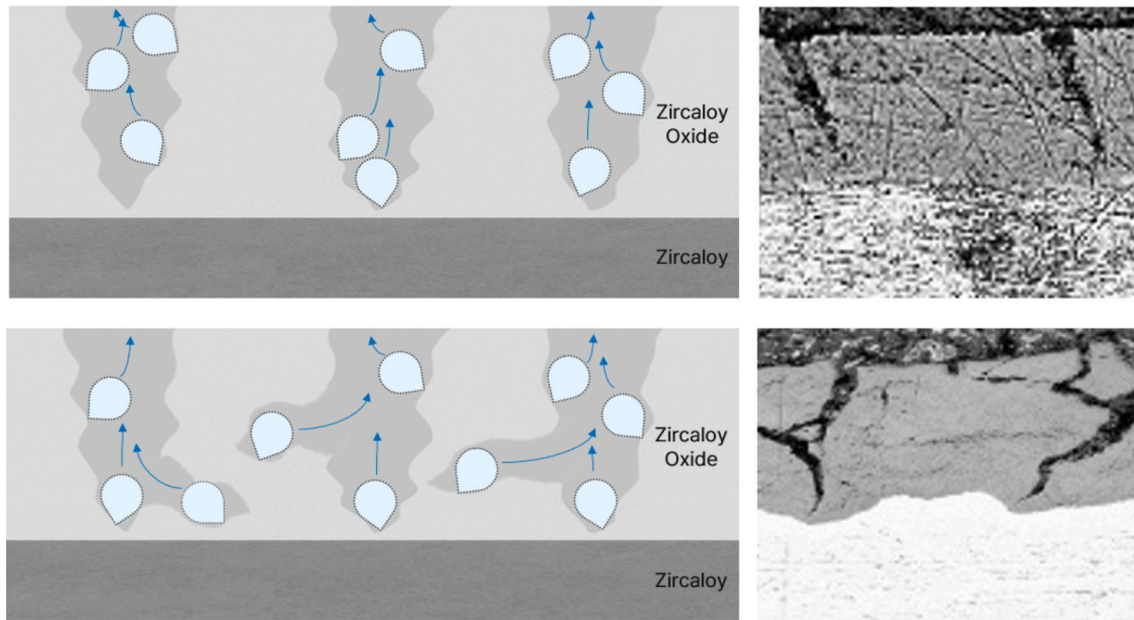


Fig. 7. (a) Bubble dynamics in specimens with transverse cracks compared with specimen oxidized at 600 °C. (b) Bubble dynamics in specimens with both transverse and axial cracks compared with specimen oxidized at 700 °C.

Table 6

Optical microscope cross-section images of specific specimens.

Oxidation temperature	600 °C	700 °C	800 °C
Cross-section images			
Oxide layer thickness	23.21 ± 4.19 μm	27.14 ± 3.12 μm	25.61 ± 5.14 μm
Contact angle	20.9 ± 4.5°	12.4 ± 2.1°	26.2 ± 4.3°
Max. size of crack in the x and y directions	x: 9.2 μm, y: 19.5 μm	x: 4.7 μm, y: 22.7 μm	x: 5.1 μm, y: 32.7 μm
CHF ratio	1.47	1.69	1.91
Cross-section images			
Oxide layer thickness	51.55 ± 3.65 μm	51.27 ± 4.87 μm	51.54 ± 6.32 μm
Contact angle	16.7 ± 4.3°	4.3 ± 1.7°	0°
Max. size of crack in the x and y directions	x: 4.2 μm, y: 38.1 μm	x: 16.8 μm, y: 46.1 μm	x: 20.7 μm, y: 67.8 μm
CHF ratio	1.01	1.34	0.93
Cross-section images			
Oxide layer thickness	77.27 ± 3.49 μm	78.09 ± 3.89 μm	73.77 ± 6.74 μm
Contact angle	0°	0°	0°
Max. size of crack in the x and y directions	x: 6.8 μm, y: 59.8 μm	x: 27.8 μm, y: 61.5 μm	x: 42.5 μm, y: 97.9 μm
Contact angle	0.75	0.71	1.22

mm), Zircaloy-4, and the oxide layer of Zircaloy-4, respectively.

The conduction resistance, which is the amount of resistance to heat flow through conduction, is defined by the thermal conductivity and geometric parameters, as shown in the following equation:

$$R_{\text{cond}} = \frac{t_{\text{Zr}}}{k_{\text{Zr}}A} + \frac{2t_{\text{ZrO}_2}}{k_{\text{ZrO}_2}A} \quad (10)$$

where R_{cond} is the conduction resistance, and A is the area of the specimen heated area, 15 (length) × 15 (width) mm. The equation shows that due to the large thermal conductivity difference between the oxidized (1.5 W/m·K) and original Zircaloy-4 (16.7 W/m·K), increasing the oxide layer thickness at the same total specimen thickness results in increased conduction resistance, and if the total specimen thickness is

Table 7

Bubble departure time of different specimens.

	Time (ms)	0 (nucleation)	8.8	17.6	26.4	35.2	44.0 (detach)
Oxidized at 600 °C for 80 h (at 27.75 kW/m ²)	High-speed image						
	Time (ms)	0 (nucleation)	8.8	17.6	26.4 (detach)	35.2	44.0
Oxidized at 800 °C for 120 min (at 34.09 kW/m ²)	High-speed image						
	Time (ms)	0 (nucleation)	8.8	17.6	26.4	35.2 (detach)	44.0
Oxidized at 800 °C for 160 min (at 31.78 kW/m ²)	High-speed image						
	Time (ms)	0 (nucleation)	8.8	17.6	26.4	35.2 (detach)	44.0

increased at the same oxide layer thickness, the conduction resistance is also increased, but not as much as when increasing the oxide layer thickness.

The oxide layer thickness, thermal conductivity, conduction resistance and CHF of the specimens are arranged in Table 8. The data shows that the CHF ratio, which represents CHF enhancement rate, exhibited decreasing trends with decreasing thermal conductivity and increasing conduction resistance, after peaking at oxide layer thicknesses between 20 and 30 μm . The conduction resistance shows constantly increasing trends with the increasing oxide layer thickness, so altering the total specimen thickness would primarily shift the magnitude of the conduction resistance without changing its monotonic trend, and has minimal influence on the optimal oxide layer thickness governed by other parameters. Although thermal conductivity and conduction resistance are not the only parameters affecting boiling heat transfer performance, the experimental data shows that they have an important role in specimens where the percentage of oxide layer to the specimen thickness is higher.

In the previous study by An et al. [40] which involved the boiling experiments of specimens with various thermal conductivities and wettabilities at the same specimen thickness, it was observed that the conduction resistance of the specimen had a significant influence on the boiling heat transfer performance. Specimens with high thermal conductivity and accordingly, low conduction resistance enable efficient heat transfer from the heater to the surface through conduction. In this regard, the conduction resistance affects the time of reactivation of nucleation sites. Snapshots in Table 7 show that the bubble departure time is higher for specimens with thicker oxide layers which have lower thermal conductivity, reinforcing this analysis. In the previous study by Zhang et al. [41], researchers observed that for isolated bubble regimes, the bubble departure frequency is mainly dominated by the bubble

nucleation time, during which phase change heat transfer through microlayer evaporation takes place. Future research aimed at clarifying this mechanism is important in understanding the fundamentals physics of boiling heat transfer.

5. Conclusion

In this study, pool boiling experiments were conducted on various flat plate type Zircaloy-4 specimens which were oxidized at the temperatures of 600 °C, 700 °C, and 800 °C for different time periods, to investigate the effects of the surface characteristics of oxidized Zircaloy-4 on boiling heat transfer performance.

1. The oxidation process creates microstructural defects at the Zircaloy-4 specimen surface, which induce capillary forces and facilitate the movement of water towards the surface. During pool boiling, this phenomenon prevents the formation of vapor films on the surface and delays the onset of CHF, enhancing heat transfer performance. However, the results from the pool boiling experiments showed that the highest CHF and HTC values were focused at between oxide layer thicknesses 20 and 30 μm , where the CHF increased by up to 147% for 600 °C, 168% for 700 °C, and 191% for 800 °C oxidized specimens compared to the bare specimen. This shows that while surface wettability greatly affected heat transfer performance in specimens with oxide layer thicknesses below 30 μm , it was less the case for specimens with higher oxide layer thicknesses.
2. Specimens oxidized at 800 °C exhibit lower crack density, surface roughness and wettability, but possess noticeable surface protrusions and wider, visually prominent cracks. These specimens demonstrate relatively higher CHF and HTC values compared to those oxidized at 600 °C and 700 °C, which suggests that across different oxidation

Table 8

Thermal conductivity, conduction resistance and CHF enhancement ratio of the specimens.

600 °C				700 °C				800 °C			
Oxide layer thickness (μm)	Thermal conductivity (W/m-K)	Conduction resistance (K/W)	CHF ratio	Oxide layer thickness (μm)	Thermal conductivity (W/m-K)	Conduction resistance (K/W)	CHF ratio	Oxide layer thickness (μm)	Thermal conductivity (W/m-K)	Conduction resistance (K/W)	CHF ratio
11	11.55	0.19	1.35	14	10.65	0.21	1.38	15	10.39	0.21	1.36
23	8.64	0.26	1.47	27	7.98	0.28	1.69	25	8.30	0.27	1.91
31	7.40	0.30	0.99	41	6.27	0.35	1.49	30	7.54	0.29	1.29
40	6.37	0.35	1.07	51	5.45	0.41	1.34	33	7.14	0.31	1.55
51	5.45	0.41	1.01	66	4.54	0.49	1.32	47	5.75	0.39	1.14
61	4.81	0.46	0.61	78	4.01	0.55	0.71	51	5.45	0.41	0.93
71	4.31	0.52	0.94	92	3.53	0.63	0.74	66	4.50	0.49	1.02
77	4.05	0.55	0.75	145	2.43	0.92	0.58	74	4.18	0.53	1.22
90	3.59	0.62	0.96								
101	3.28	0.68	0.62								

temperatures, surface characteristics other than roughness and wettability, such as crack morphology and surface topology, have a more significant influence on boiling heat transfer performance. The high-speed images of the bubble dynamics taken from the experiments further support this theory, where the generation of bubbles on the surface was weaker for the 600 °C oxidized specimen than for the 800 °C oxidized specimen. Narrower cracks generate higher capillary forces in theory, but they also have a higher tendency to merge and hinder bubble escape, so their effect on boiling heat transfer depends on whether they provide effective liquid replenishment without blocking vapor removal.

3. The structural orientation of the cracks within the oxide layer have two-sided effects on boiling heat transfer performance. While vertical cracks enable the capillary wicking effect and draw the cooling fluid towards the heated surface, horizontal cracks complicate the vapor bubble escape. This phenomenon obstructs the replenishment of the cooling fluid, which in turn hinders boiling heat transfer performance. The cracks in the horizontal direction, which diverged from the vertical cracks exposed to the surface, were more developed for higher oxide layer thicknesses in 700 °C and 800 °C oxidized specimens, but not in 600 °C. This characteristic is one of the driving forces behind the deterioration of heat transfer performance at high oxide layer thicknesses for specimens oxidized at the same temperature.
4. The thermal conductivity of Zircaloy-4 oxide is significantly smaller than that of the original Zircaloy-4, leading to the significantly reduced overall thermal conductivity of specimens with thicker oxide layers. The thermal conductivity and the corresponding conduction resistance of the specimen significantly affects boiling heat transfer performance by influencing the bubble departure time. In this context, the CHF ratio exhibited decreasing trend with decreasing thermal conductivity and increasing conduction resistance, after peaking at oxide layer thicknesses between 20 and 30 μm . The experimental data shows that although thermal conductivity and conduction resistance are not the only parameters affecting heat transfer performance, they have an important role in determining the boiling heat transfer performance of specimens where the proportion of oxide layer to the total specimen thickness is higher.
5. Specimens with thicker oxide layers possess lower thermal conductivities, and more developed horizontal cracks with more complicated bubble escape routes, affecting boiling heat transfer negatively. On the other hand, surface microstructures with certain conditions such as vertical direction cracks enhance boiling heat transfer by inducing the capillary wicking effect, which facilitates the rewetting of the heated surface. The positive effects on boiling heat transfer by these surface microstructures are most influential at the oxide layer thicknesses between 20 and 30 μm , and it still continues to be dominant until the thicknesses between 50 and 60 μm . Beyond this range, the negative effects by the decreasing thermal conductivity and bubble escape complexity outweigh the positive effects by the surface microstructures, leading to deterioration in boiling heat transfer performance relative to the bare specimen. However, this trend is less evident in the 800 °C specimens, likely due to their distinct surface characteristics.

CRedit authorship contribution statement

Sung Jin Kim: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Sangmin Song:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **Junbeom Song:** Data curation, Formal analysis. **Il Woong Park:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Hee Sang Yoo:** Writing – review & editing. **Hyun Sun Park:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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References

- [1] C.D.F. Azevedo, Selection of fuel cladding material for nuclear fission reactors, *Eng. Fail. Anal.* 18 (8) (2011) 1943–1962.
- [2] P. Yvon, F. Carré, Structural materials challenges for advanced reactor systems, *J. Nucl. Mater.* 385 (2) (2009) 217–222.
- [3] G.P. Sabol, R.J. Comstock, R.A. Weiner, P. Larouere, R.N. Stanutz, In-reactor corrosion performance of ZIRLO™ and zircaloy-4, in: *Zirconium in the nuclear industry: Tenth International Symposium*, ASTM International, 1994.
- [4] K.J. Annand, The nanoscale mechanisms of Zircaloy-4 corrosion in simulated nuclear reactor conditions (Doctoral dissertation, University of Glasgow), 2018.
- [5] T. Allen, H. Burlet, R.K. Nanstad, M. Samaras, S. Ukai, Advanced structural materials and cladding, *MRS Bull.* 34 (1) (2009) 20–27.
- [6] Q. Li, Q.J. Kang, M.M. Francois, Y.L. He, K.H. Luo, Lattice boltzmann modeling of boiling heat transfer: the boiling curve and the effects of wettability, *Int. J. Heat Mass Transf.* 85 (2015) 787–796.
- [7] W.M. Rohsenow, A method of correlating heat-transfer data for surface boiling of liquids, *J. Fluids Eng.* 74 (6) (1952) 969–975.
- [8] S. Rashidi, F. Hormozi, M.M. Sarafraz, Potentials of boiling heat transfer in advanced thermal energy systems: S. rashidi et al, *J. Therm. Anal. Calorim.* 143 (3) (2021) 1833–1854.
- [9] H.J. Cho, D.J. Preston, Y. Zhu, E.N. Wang, Nanoengineered materials for liquid–vapour phase-change heat transfer, *Nat. Rev. Mater.* 2 (2) (2016) 1–17.
- [10] J. Straub, Bubble—bubbles—boiling, *Microgravity-Sci. Technol.* 16 (1) (2005) 242–248.
- [11] J. Kim, Review of nucleate pool boiling bubble heat transfer mechanisms, *Int. J. Multiphase Flow* 35 (12) (2009) 1067–1076.
- [12] D. Wang, T. Lin, X. Quan, Enhanced liquid replenishment for pool boiling heat transfer and its CHF mechanism on patterned freeze-casted surfaces, *Exp. Therm. Fluid Sci.* 150 (2024) 111035.
- [13] Y.S. Choi, S.J. Kim, I.W. Park, H.S. Park, Y.G. Lee, Experimental investigation on the influence of copper foam characteristics on pool boiling heat transfer, *Int. Commun. Heat Mass Transf.* 159 (2024) 108318.
- [14] B. Lee, Y.S. Na, S.C. Jang, S.J. Kim, Analysis of severe accident by beam tube break in 30 MW pool type research reactor with U3Si Fuel, *Nucl. Eng. Technol.* (2025) 103956.
- [15] J.S. Moya, M. Diaz, J.F. Bartolome, E. Roman, J.L. Sacedon, J. Izquierdo, Zirconium oxide film formation on zircaloy by water corrosion, *Acta Mater.* 48 (18–19) (2000) 4749–4754.
- [16] J. Hu, B. Setiadinata, T. Aarholt, A. Garner, A. Vilalta-Clemente, J. Partezana, C. Grovenor, Understanding Corrosion and Hydrogen Pickup of Zirconium Fuel Cladding Alloys: The Role of Oxide Microstructure. Porosity, Suboxides, and Second-Phase Particles, in: *Zirconium in the Nuclear Industry*, 2018, pp. 93–126.
- [17] N.V. Upot, K.F. Rabbi, S. Khodakarami, J.Y. Ho, J.K. Mendizabal, N. Miljkovic, Advances in micro and nanoengineered surfaces for enhancing boiling and condensation heat transfer: a review, *Nanoscale Adv.* 5 (5) (2023) 1232–1270.
- [18] K. Svanholm, M.P. Breggi, F. D'Auria, R. Ianiri, Halden reactors IFA-511.2 and IFA-54x: experimental series under adverse core cooling conditions, *Exp. Therm. Fluid Sci.* 11 (1) (1995) 77–100.
- [19] N. Kim, H.H. Son, S.J. Kim, Oxidation effect on pool boiling critical heat flux enhancement of Cr-coated surface for accident-tolerant fuel cladding application, *Int. J. Heat Mass Transf.* 144 (2019) 118655.
- [20] H. Maki, Heat transfer characteristics of zircaloy-2 oxide film, *J. Nucl. Sci. Technol.* 10 (3) (1973) 170–175.
- [21] J. Lee, H. Jeong, Y. Bang, Thermal resistance effects of crud and oxide layers to the safety analysis, *Topfuel, Prague Czech2018* (2018).
- [22] C.Y. Lee, T.H. Chun, W. Kee In, Critical heat flux of oxidized zircaloy surface in saturated water pool boiling, *J. Nucl. Sci. Technol.* 52 (4) (2015) 596–606.
- [23] G.C. Lee, H. Noh, H. Yeom, H. Jo, T.K. Kim, M. Kim, H.S. Park, Zirconium-silicide coating on zircaloy-4 substrate for accident tolerance: effects on oxidation resistance and boiling, *Ann. Nucl. Energy* 126 (2019) 350–358.

- [24] H. Yeom, E. Gutierrez, H. Jo, Y. Zhou, K. Mondry, K. Sridharan, M. Corradini, Pool boiling critical heat flux studies of accident tolerant fuel cladding materials, *Nucl. Eng. Des.* 370 (2020) 110919.
- [25] M. He, A.F. Ali, M. Chen, Steady-state pool boiling heat transfer experimental studies of horizontally-placed tubes, *Int. J. Heat Mass Transf.* 196 (2022) 123270.
- [26] H.S. Ahn, C. Lee, H. Kim, H. Jo, S. Kang, J. Kim, M.H. Kim, Pool boiling CHF enhancement by micro/nanoscale modification of zircaloy-4 surface, *Nucl. Eng. Des.* 240 (10) (2010) 3350–3360.
- [27] T.A. Moreira, D. Lee, M.H. Anderson, Critical heat flux on zircaloy and accident tolerant fuel cladding under prototypical conditions of pressurized and boiling water reactors, *Appl. Therm. Eng.* 213 (2022) 118740.
- [28] J.H. Bae, S.J. Hong, T.S. Ro, Effect of chromium coated zircaloy cladding of accident tolerant fuel on the best estimate LOCA methodology, *Prog. Nucl. Energy* 151 (2022) 104330.
- [29] T.A. Moreira, K.D. Murray, M.E. Conner, Y. Sung, J. Walters, B.R. Maier, M. H. Anderson, Time in DNB experimental study on Cr coated zircaloy cladding, *Appl. Therm. Eng.* 248 (2024) 123266.
- [30] K. Dong, S. He, S. Li, D. Chen, J. Zhao, Experimental study of roughness effect on critical heat flux in subcooled flow boiling, *Int. J. Heat Mass Transf.* 255 (2026) 127806.
- [31] J.H. Seong, C. Wang, B. Phillips, M. Bucci, Separate effect of oxidation on the subcooled flow boiling performance of Zircaloy-4 at atmospheric pressure, *Int. J. Heat Mass Transf.* 188 (2022) 122620.
- [32] S.J. Kline, Describing uncertainties in single-sample experiments, *Mech. Eng.* 75 (1963) 3–8.
- [33] N. Zuber, Hydrodynamic aspects of boiling heat transfer (No. 4439), United States Atomic Energy Commission, Technical Information Service, 1959.
- [34] G. Liang, I. Mudawar, Review of pool boiling enhancement by surface modification, *Int. J. Heat Mass Transf.* 128 (2019) 892–933.
- [35] S.M. Muthu, H.B. Lee, B.O. Okonkwo, D. Wang, C. Jang, T. Na, Accelerated formation of oxide layers on Zircaloy-4 utilizing air oxidation and comparison with water-corroded oxide layers, *Materials* 16 (24) (2023) 7589.
- [36] T. Sawabe, T. Sonoda, M. Furuya, S. Kitajima, H. Takano, Residual stress distribution in oxide films formed on Zircaloy-2, *J. Nucl. Mater.* 466 (2015) 658–665.
- [37] H.S. Ahn, C. Lee, J. Kim, M.H. Kim, The effect of capillary wicking action of micro/nano structures on pool boiling critical heat flux, *Int. J. Heat Mass Transf.* 55 (1–3) (2012) 89–92.
- [38] C.Y. Lee, T.H. Chun, W. Kee In, Critical heat flux of oxidized zircaloy surface in saturated water pool boiling, *J. Nucl. Sci. Technol.* 52 (4) (2015) 596–606.
- [39] E. Akbari, S.G. Holagh, H. Saffari, M. Shafiee, On the effect of silver nanoparticles deposition on porous copper foams on pool boiling heat transfer enhancement: an experimental visualization, *Heat Mass Transf.* 58 (3) (2022) 447–466.
- [40] Y. An, C. Huang, X. Wang, Effects of thermal conductivity and wettability of porous materials on the boiling heat transfer, *Int. J. Therm. Sci.* 170 (2021) 107110.
- [41] L. Zhang, S. Gong, Z. Lu, P. Cheng, E.N. Wang, A unified relationship between bubble departure frequency and diameter during saturated nucleate pool boiling, *Int. J. Heat Mass Transf.* 165 (2021) 120640.